Tecnoforge
FITTINGS & FLANGES
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Società con Unico Socio e soggetta alla direzione
e coordinamento di Valvitalia Finanziaria S.p.A.
Cap. Soc. € 130.000.000,00 i.v.
C.F. / P.IVA / n. Registro Imprese 05962420963
R.E.A. 261465 - Pavia

VALVITALIA

Doc.: INSP.REPORT Rev. 1

Inspection Certificate 134556

BS - EN 10204-2004 3.1

Rev 01

Date 07/07/2016

Job TF	:	V16-000052
Purchaser	:	SYNERGY INTERNATIONAL METALS PTE LTD
Order	:	PO-0116-1383 REV. 02
Specification	:	MSS-SP-75 Ed.2014
Construction Standard	:	ASME B16.9/12 - B16.25/12

Item Y/R	Item O/R	Quantity	Descriptions	Heat	Lot
45	45	75	CAP 10" XS WPHY52	761230	S560H

Type of Construction	:	SEAMLESS PRODUCED BY PLATE
Method of Fabrication	:	HOT FORMING BETWEEN 780°-980° COOLING IN STILL AIR
Heat Treatment	:	NORMALIZING 920°C COOLING IN STILL AIR
Mill Test Report N.	:	Q0414314
Supplier	:	VOESTALPINE GROBBLECH GmbH

Ladle Analysis

Item Y/R	Item O/R	Heat	C %	Mn %	Si %	P %	S %	Cr %	Mo %	Ni %	Cu %	Al %	V %	Nb %	B %	Ti %
45	45	761230	0.164	1.170	0.383	0.007	0.0007	0.123	0.035	0.270	0.156	0.034	0.004	0.0170	0.0002	0.0010
	Product		0.164	1.150	0.379	0.007	0.0009	0.125	0.036	0.270	0.159	0.032	0.004	0.0180	0.0002	0.0020

$CE1 = C + \frac{MN}{6} = 0,36$ $\quad CE2 = C + \frac{MN}{6} + 0,04 = 0,40$ $\quad CE3 = C + \frac{MN}{6} + \frac{CR + V + MO}{5} + \frac{CU + NI}{15} = 0,42$

CE1 - CE2 Ladle Analysis (%) CE3 ——— Product (if applicable) ———

Tensile Test Base Material Direction: TRANSVERSAL Weld Direction:

Item Y/R	Item O/R	Test	Heat	Y.S. MPA	T.S. MPA	Elongation % (Lo=50mm)	Y.S. MPA	T.S. MPA	Elongation % (Lo=50mm)
45	45	346824	761230	391	544	38.4	0	0	0.0

Impact Test Direction: TRANSVERSAL Temp.: -46 °C Type: KCV

Item Y/R	Item O/R	Test	Size MM.	Section CM2		Impact Value (J) - Avg (1)	(2)	(3)		Shear Area (%) - Avg (1)	(2)	(3)		Lat. Expansion (mm) - Avg (1)	(2)	(3)
45	45	346824	10 X 10	0,800	Base Mat. :	162	157	135	151	90	90	90	90			
					H.A.Z. :	0	0	0	0							
					Weld :	0	0	0	0							

Bending Test

Hardness Test Base Mat. 235 HBW MAX H.A.Z. Weld

Note

We certify that the items comply with order requirements and above mentioned specifications.
All items are marked with low stress die stamps or interrupted dot stamps.
Visual and dimensional inspection: SATISFACTORY Mechanical test carried out on finished items - (YES / [redacted])

Trade Mark	Mfr Authorized Inspector	Official Designated Inspector	Purchaser Authorized Inspector